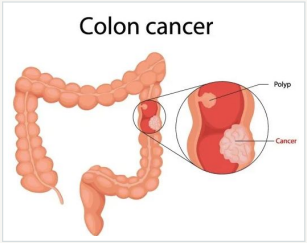
Physics-Informed Neural Network for a Reduced WNT-HOX-Retinoic Acid Model of Colorectal Cancer

Amirkhan Aidarkhan, Nathaniel Kim

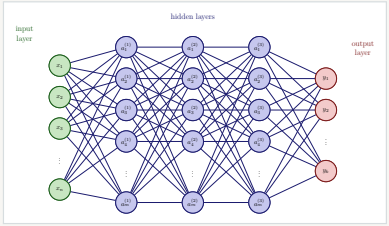
Advisor: Dr. Kataboh

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July 21, 2026



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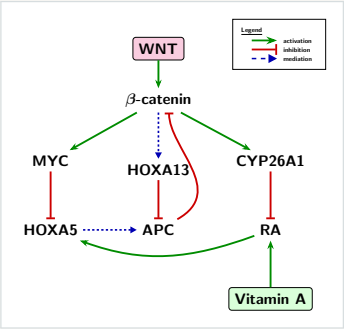
Overview

- Biology/Schematic
- Model
- Forward PINN
- Inverse PINN
- Inverse Bayesian PINN
- Next Steps

Biological Network Governing WNT–HOX–RA Signaling

Objective

Develop a reduced mathematical model describing the interactions among the WNT, HOX, and Retinoic Acid (RA) pathways during colorectal cancer progression.



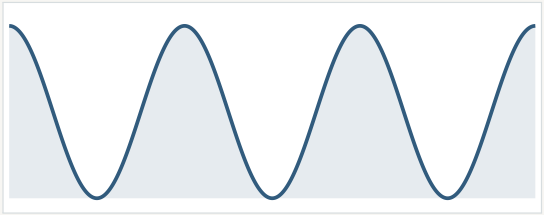
State Variables

$$\mathbf{y} = [B(t), P(t), H_5(t), H_{13}(t), M(t), R(t), C(t)]$$

- B : β-catenin
- P : APC
- H_5 : HOXA5
- H_{13} : HOXA13
- M : MYC
- R : Retinoic Acid
- C : CYP26A1

RA Input + Treatment

RA Input



$$A_d\left[1+\cos\left(\frac{2\pi t}{T_d}-\phi\right)\right]$$

ATRA Treatment



$$u_R(t)=\frac{D}{2}[\tanh q(t-t_1)-\tanh q(t-t_2)]$$

Dimensional Model

$$\frac{dB}{dt} = a_B W + v_{13}\frac{H_{13}^{n_H}}{K_{13}^{n_H} + H_{13}^{n_H}} - d_B B - k_{PB} P B - v_5 \frac{H_5 B}{K_5 + B}$$

$$\frac{dP}{dt} = a_P \frac{1 + \rho_5 H_5}{1 + \rho_B B + \rho_{13} H_{13}} - \Big(d_{P0} + d_{P1}(1 - \Theta_P)\Big) P$$

$$\frac{dH_5}{dt} = a_5 + v_R \frac{R}{K_R + R} - d_5 H_5 - v_M \frac{M H_5}{K_M + M}$$

$$\frac{dH_{13}}{dt} = a_{13} + v_{B13}\frac{B^{n_B}}{K_{B13}^{n_B} + B^{n_B}} + v_{M13}\frac{M^{n_M}}{K_{M13}^{n_M} + M^{n_M}} - d_{13} H_{13}$$

$$\frac{dM}{dt} = a_M + v_{BM}\frac{B^{n_B}}{K_{BM}^{n_B} + B^{n_B}} - d_M M$$

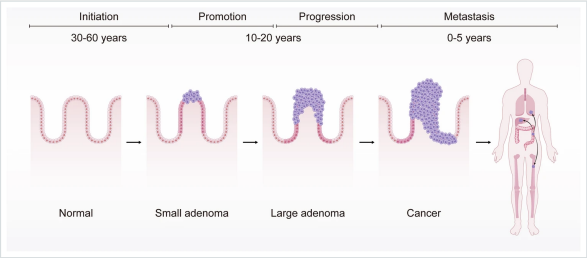
$$\frac{dR}{dt} = u_0 + A_d\left[1 + \cos\left(\frac{2\pi t}{T_d} - \phi\right)\right] + \frac{D}{2}\Big[\tanh(q(t - t_1)) - \tanh(q(t - t_2))\Big] - d_R R - k_{CR} C R$$

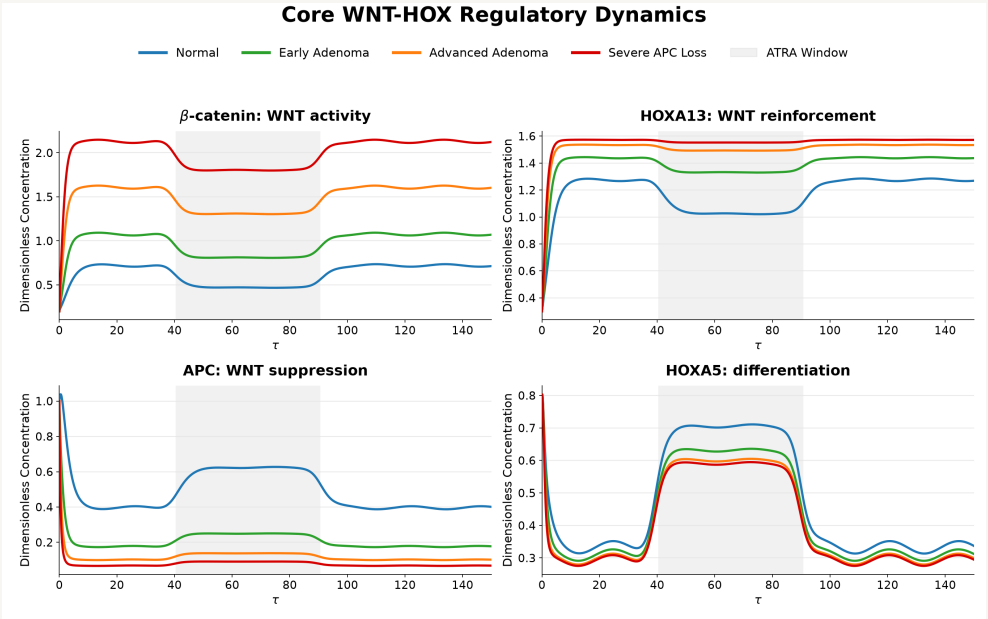
$$\frac{dC}{dt} = a_C + v_{RC}\frac{R}{K_{RC} + R} + v_{BC}\frac{B^{n_B}}{K_{BC}^{n_B} + B^{n_B}} - d_C C$$

4 Regimes

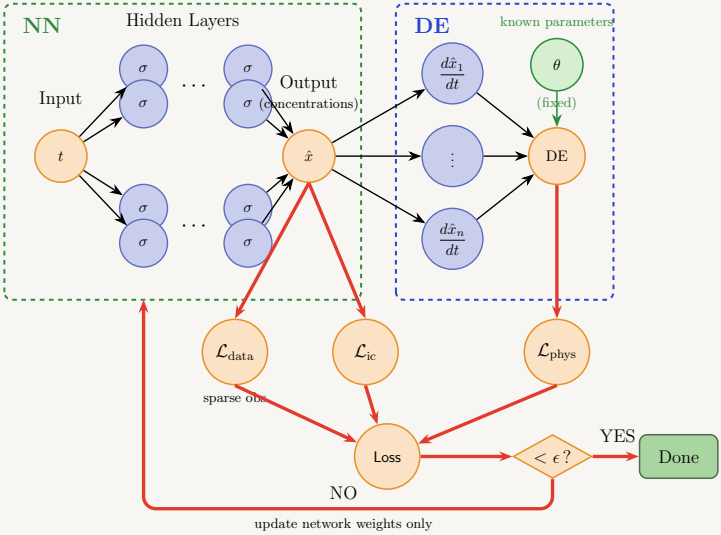
Disease progression is modeled by increasing WNT signaling (W) and progressive loss of APC function (θ_P), mimicking CRC progression.

- “Normal”
  - $W = 0.8$, $\theta_P = 1.0$
- “Early Adenoma”
  - $W = 1.0$, $\theta_P = 0.75$
- “Advanced Adenoma”
  - $W = 1.5$, $\theta_P = 0.5$
- “Severe APC Loss”
  - $W = 2.0$, $\theta_P = 0.25$

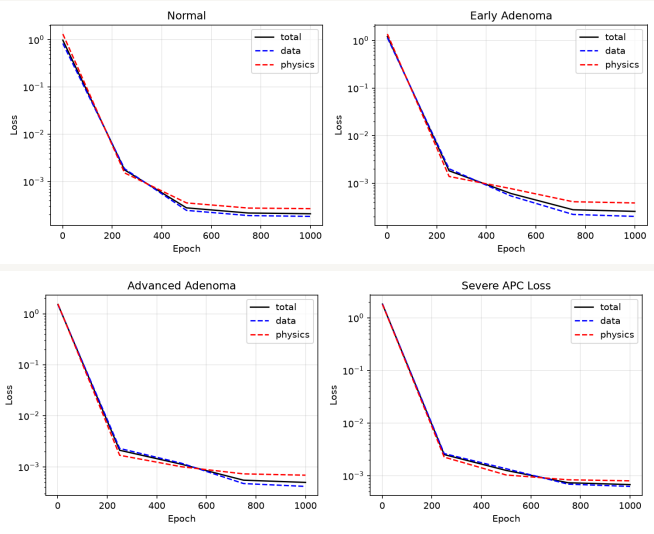


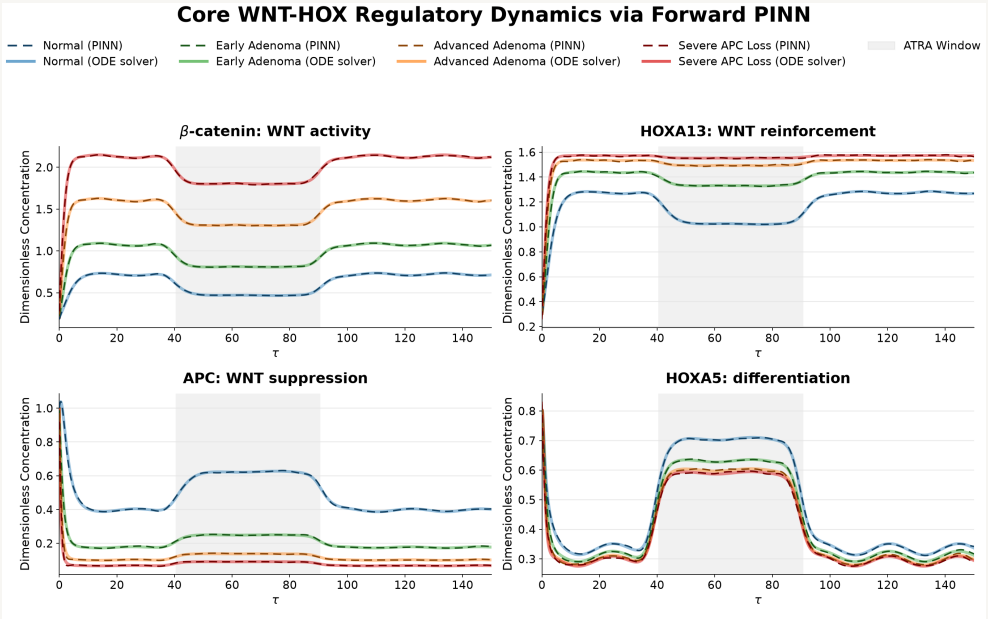

Core WNT-HOX Regulatory Dynamics
Normal
Early Adenoma
Advanced Adenoma
Severe APC Loss
ATRA Window
β-catenin: WNT activity
HOXA13: WNT reinforcement
APC: WNT suppression
HOXA5: differentiation
Dimensionless Concentration
τ

Forward PINN

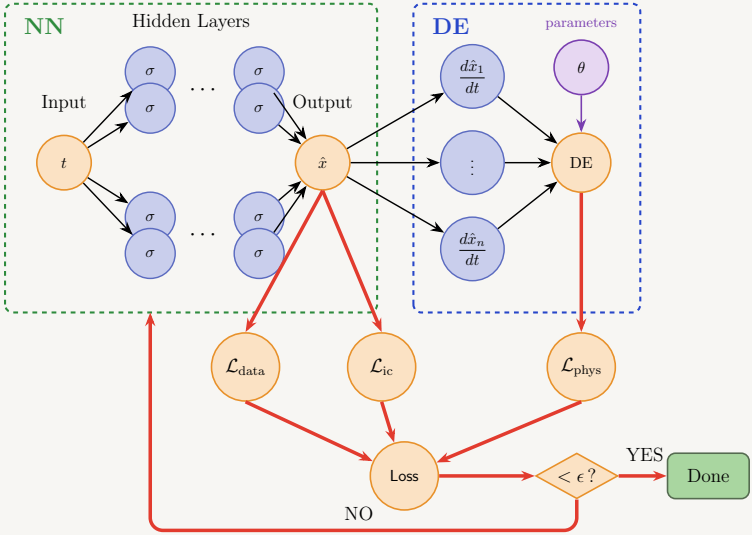


Forward PINN Loss

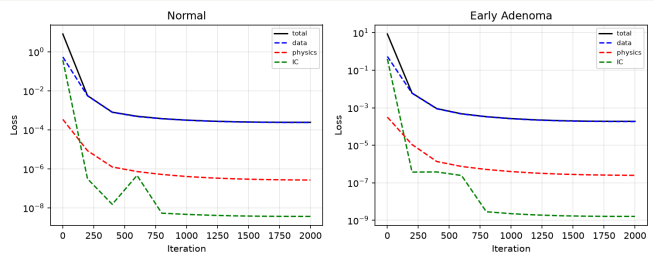


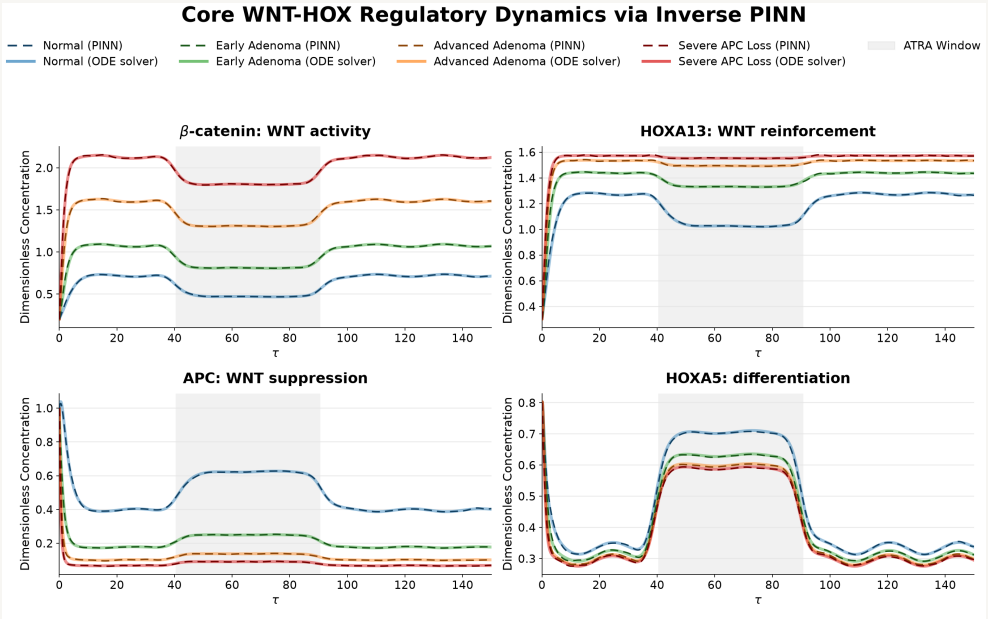

Core WNT-HOX Regulatory Dynamics via Forward PINN
Normal (PINN)
Normal (ODE solver)
Early Adenoma (PINN)
Early Adenoma (ODE solver)
Advanced Adenoma (PINN)
Advanced Adenoma (ODE solver)
Severe APC Loss (PINN)
Severe APC Loss (ODE solver)
ATRA Window
β-catenin: WNT activity
HOXA13: WNT reinforcement
APC: WNT suppression
HOXA5: differentiation
Dimensionless Concentration
τ

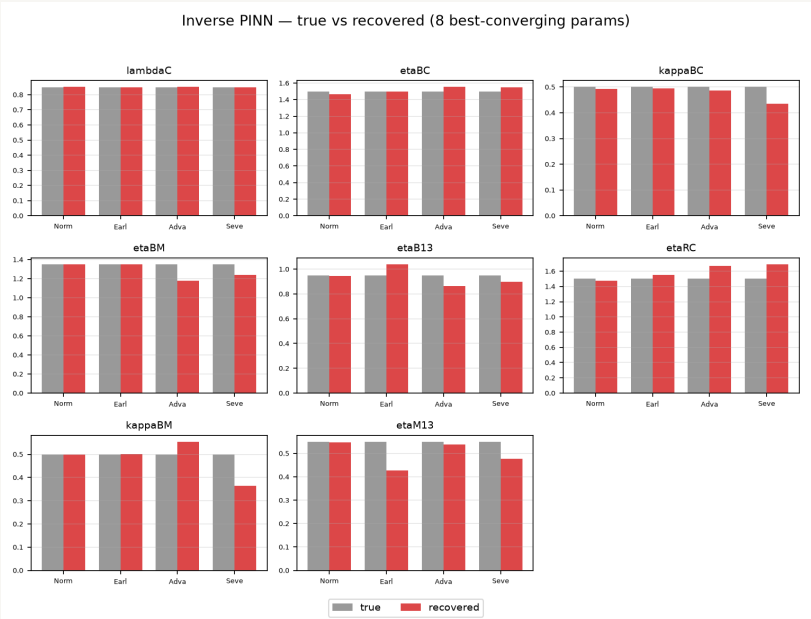
Inverse PINN



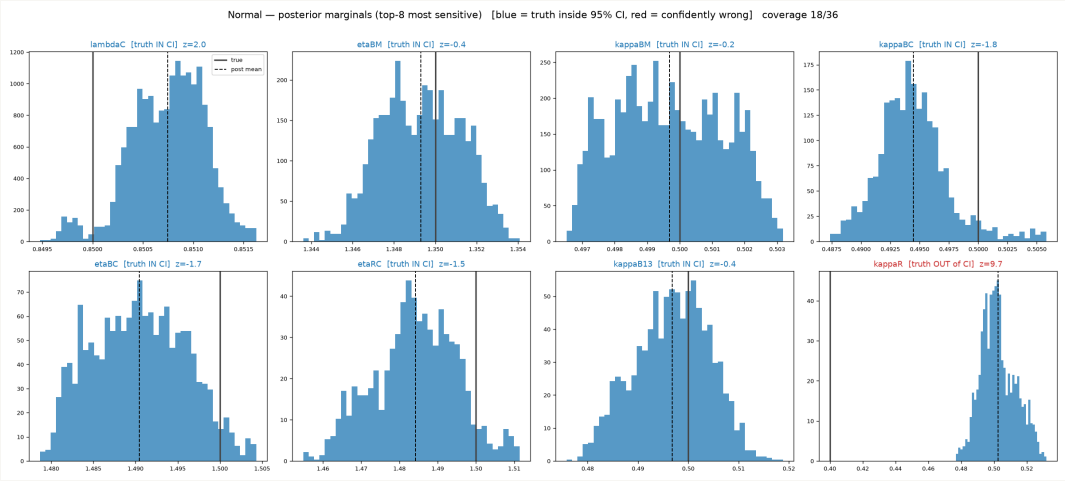
Inverse PINN Loss



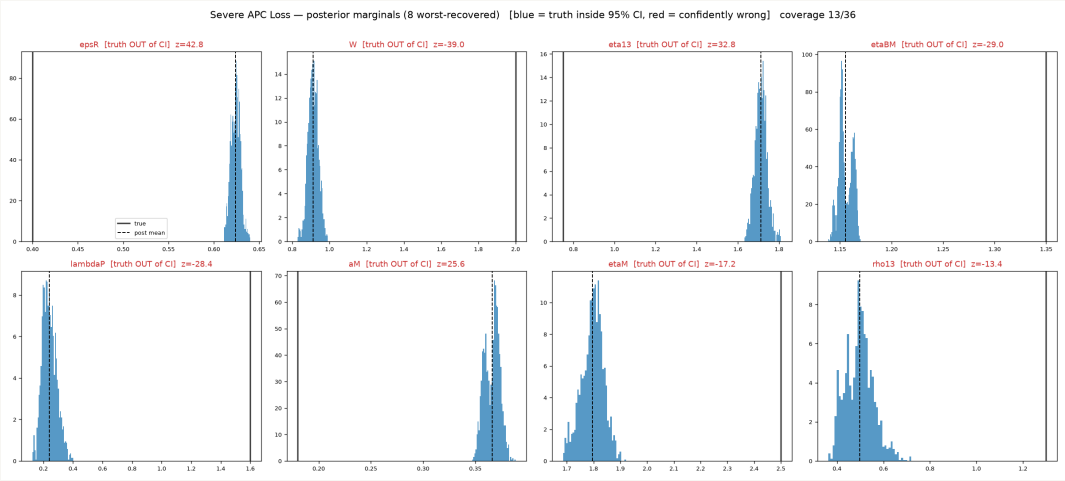

Core WNT-HOX Regulatory Dynamics via Inverse PINN
Normal (PINN)
Normal (ODE solver)
Early Adenoma (PINN)
Early Adenoma (ODE solver)
Advanced Adenoma (PINN)
Advanced Adenoma (ODE solver)
Severe APC Loss (PINN)
Severe APC Loss (ODE solver)
ATRA Window
β-catenin: WNT activity
HOXA13: WNT reinforcement
APC: WNT suppression
HOXA5: differentiation
Dimensionless Concentration
τ


Inverse PINN — true vs recovered (8 best-converging params)
lambdaC
etaBC
kappaBC
etaBM
etaB13
etaRC
kappaBM
etaM13
Norm
Earl
Adva
Seve
true
recovered

Inverse Bayesian PINN - uncertainty quantification



Inverse Bayesian PINN - continued



Parameter Recovery Comparison

	Bayesian inverse PINN	Inverse PINN	ODE-fit
“Normal”	**19**	17	18/36
“Early Adenoma”	**20**	16	17/36
“Advanced Adenoma”	11	10	**13/36**
“Severe APC Loss”	8	7	**11/36**

Next Steps

- Improve Bayesian PINN
  - Investigate limitations
- Hybrid Mechanistic-Neural Model
- Stochastic Modeling



Key References

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Thank you!

Questions?

Thank you very much to the **Provost's Office** and the **Academic Divisions** for funding our summer research project!